

Wireless Signal Classification via Deep Learning (Generative Adversarial, Open Set Incremental and Dual Head)

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Abstract

Wireless signal environments are becoming increasingly crowded, with diverse and dynamic signal types competing for bandwidth. Identifying and classifying these signals in real-time is critical for a wide range of applications, including spectrum management, interference reduction, and national security. Existing signal classification techniques, while effective in controlled environments, often struggle to generalize across different datasets and signal types. Moreover, new and unknown signal types regularly emerge, posing a challenge for traditional supervised learning models. We address these challenges by developing a dual-head network reinforced with Generative Adversarial Networks (GANs) and exploring open set incremental learning to improve the generalization and adaptability of signal classification models.

Publicly available datasets from DeepSig, RADIOML 2018.01A and RADIOML 2016.10A [3] are used for the purpose of comparison of prior related works. The goal is to classify the modulation type of signals and build on performance from previous work using machine learning¹. The work done here shows that we can get state of the art max accuracy of 92% with a 30% increase in inference time. This is better than any prior reproducible work to our knowledge.

1. Introduction

Radio Frequency (RF) signals are a difficult dataset to classify for multiple reasons. AI and ML have focused heavily on images because humans have a propensity to be intrigued by things they can see. Signals also have a temporal aspect that have rapid shifts in phase and amplitude at the gigahertz and beyond scale. Temporal rapid shifts make it difficult to apply image based algorithms which could mask minor pixel issues and still have prominent results. The DeepSig authors note numerous errata in their dataset and therefore it is not realistic to expect 100% accuracy. However some related

work does exists that we can review, compare, and contrast with.

In RF applications, a two-dimensional sinusoidal signal is sampled with a complex component, adding a third dimension. With time on one axis, the in-phase and quadrature-phase components are known as the I/Q data points. The formulas below give them, where A is the amplitude, f is the frequency, and ϕ is the phase of time, t.

$$I(t) = A * \cos(2\pi ft + \phi)$$

$$Q(t) = A * \sin(2\pi ft + \phi)$$

Digital signals can be uniquely identified by their modulation scheme. Labeled I/Q data can be used to classify a signal based on a set of features (i.e., frequency, bandwidth, power, etc.) using supervised and unsupervised machine-learning techniques. The RML2016.10A dataset is provided with eleven modulation classifications, including FM, AM x 2, and 8 digitally modulated signals (CPFSK, GFSK, BPSK, QPSK, 8PSK, PAM4, QAM16, QAM64). The dataset is labeled, with each modulation consisting of samples at 20 different SNR levels (-20dB to 18dB, in 2dB steps). There are one thousand samples for each SNR/Modulation combination. This provides 220,000 samples, each consisting of a 2 x 128-time series vector of I/Q data points. The RML2018 dataset contains 2.5M samples of 24 modulations (OOK, 4ASK, 8ASK, BPSK, QPSK, 8PSK, 16PSK, 32PSK, 16APSK, 32APSK, 64APSK, 128APSK, 16QAM, 32QAM, 64QAM, 128QAM, 256QAM, AM-SSB-WC, AM-SSB-SC, AM-DSB-WC, AM-DSB-SC, FM, GMSK, OQPSK), each in a 1024 x 2 vector of I/Q data. This data has SNR from -20dB to 30dB in 2dB steps. We assume our data is independent and identically distributed (i.i.d); all our data come from the same distribution. The data requires no cleanup; duplicates do not exist since DeepSig provides it clean.

From prior work¹, we hypothesis that the relatively small RML2016.10A dataset, with 20k samples per modulation is a major contributor to the accuracy of our model. Signal to Noise (SNR) ratios below 0dB are naturally tough to decode and this reduces the sample set almost in half. Furthermore the inherent similarities of WBFM and AM-DSB as seen in **Figure 1**, as well as QAM16 vs QAM64 as seen in **Figure 2**, with similar signal characteristics such as phase changes, modulation

¹<https://github.com/gawdygoda/ML-wireless-signal-classification/blob/main/Wireless%20Signal%20Classification%20via%20Machine%20Learning%20-%20Final.pdf>

and amplitude require additional training details for model recognition.

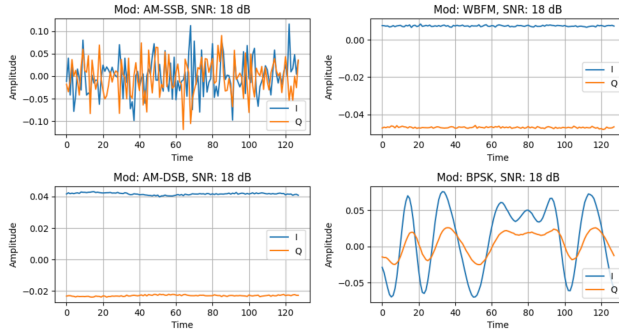


Figure 1: RML2016.10A Sample Signals

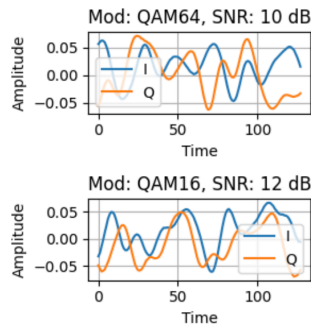


Figure 2: RML2016.10A QAM Signal Samples

With these challenges our main contributions are to use a unsupervised generative adversarial network (GAN) to create additional samples for training. Specifically, we use an auxiliary classifier GAN as we have multiple modulation types to classify, a form of a conditional GAN. Work by Odena et al. explains how an ACGAN is preferable for learning generation when your dataset can be cleanly divided into subsets [7]. In a way, it's like having an individual GAN per class, see **Figure 3**.

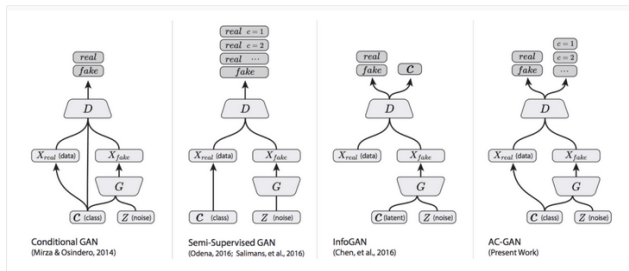


Figure 3: GAN Evolution [7]

While our dataset is synthetic, and we could generate additional samples, in the real world, data would come from over-the-air (OTA) capture. Therefore, the benefits of generating additional sample sets when only limited OTA data exists it advantageous. Additionally, we also

use the RML2018.01A dataset to help train our model. This dataset contains 4x the number of samples and 8x higher signal resolution, contributing positive effects to our small dataset challenge. Similar signals classification can also be improved with these techniques, however to combat this further, we implement the use of Open Set Incremental Learning (OSIL). To allow the model to adapt to new, previously unseen signal types it will be trained on a subset of modulations initially and will be updated incrementally as new signal types (or "novel" signals) are detected without forgetting previously learned ones. Our experiments use the same base network from our previous work and show that OSIL results in equivalent accuracy without forgetting previously known signals. Lastly, to attempt to improve accuracy, we implement a dual head model that incorporates a convolutional neural network (CNN) model that takes in frequency-domain features extracted using FFT (Fast Fourier Transform). The other head is the LSTM RNN head with the raw I/Q samples as inputs, like the original model. The idea is if we can home in the critical features the network should pay attention to via domain knowledge, it should allow for the rapid shifts in phase and amplitude to be detected.

2. Related Work

Much of the machine learning, and specifically deep learning research work has heavily focused on images, time-series signals have seen less love. However, a handful of related work does exist. We explore prior work, but take note, prior work that does not produce source code and/or enough details to reproduce the work, are discarded since we cannot validate their results.

GANs for the RFML dataset is not new. However prior work by Li et al. [2] shows a max accuracy of ~85% for an RCGAN that uses a specialized version of an ACGAN to classify the dataset. In another research by Hou et al., the authors propose a dual head LSTM RNN that takes in the raw I/Q samples as well as a slow fusion subsample of each of the I and Q datapoints. The authors achieve a high max accuracy of ~93% (91.3% when locally reproduced) [6]. These networks are heavy and do not adapt to new unknown signals as we will show with our research.

3. Technical Methods and Results

Common algorithms like convolution and max pooling are likely to lose rapidly changing information, such as instantaneous phase or amplitude. To better capture these temporal dynamics, we use natural language processing models like recurrent neural networks (RNNs). For our base model, we adopt a Long Short-Term Memory (LSTM) RNN from our previous work, which will serve as the foundation for our research. **Figure 4**, **Figure 5**, and **Figure 6** show the baseline numbers, model, and SNR plot

with a peak accuracy of ~83%. The challenges mentioned earlier are not a one-size-fits-all type problem. Four methods were explored, which we will expand on independently.

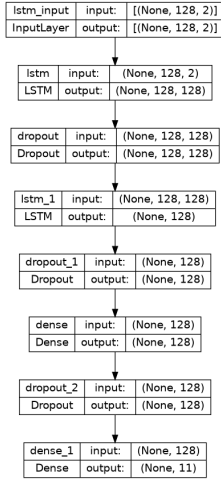


Figure 4: Base Model – Layers

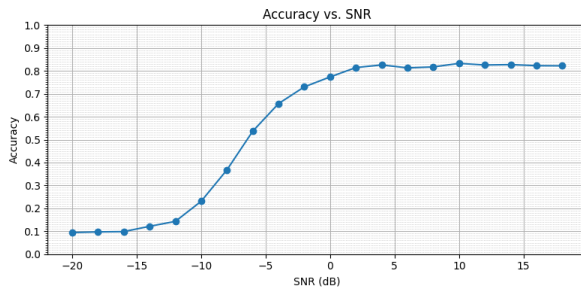


Figure 5: Base Model – SNR Accuracy Curve

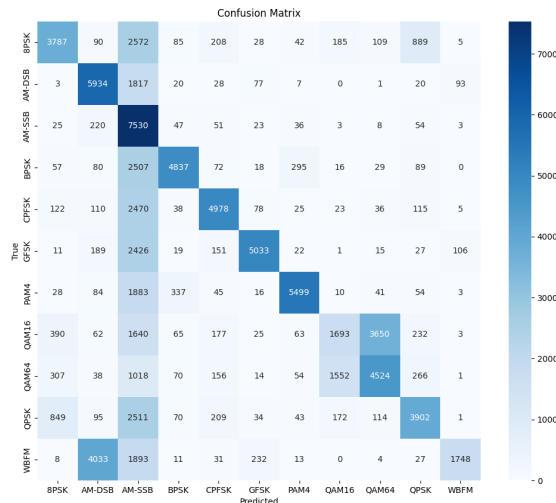


Figure 6: Base Model – Confusion Matrix

Open set incremental learning starts with training the

base model with a subset of modulations. Different scenarios were experimented with, that all had similar outcomes. We end up learning all the classes except WBFM, QAM16, QAM64, AM-DSB, and AM-SSB as those needed extra attention from the confusion matrix. Note that from prior work, we know lower SNR ($> -6\text{dB}$) signals get classified as AM-SSB as it resembles noise the closest (Figure 7). We believe this to be one of the errata

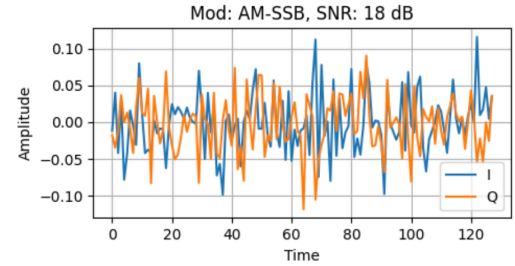


Figure 7: AM-SSB resembles noise

in the dataset. For this reason, it is also selected to be incrementally learned. Knowledge distillation was implemented to ensure we don't lose old signal classifications and can train/generalize for new signals. This involved transferring the weights from the base model to a teacher model. The base model becomes the student and is trained with a loss function that looks at the true target value as well as the predicted target value for the new incremental class. Alpha (α) is a mixing hyperparameter that sets the balance for the teacher model knows (old knowledge) and what the student model learns (new knowledge) from the new class. KL-Divergence is used in the loss function to determine the generalization of teacher model given the student model, and cross entropy is used to ensure the model also learns from the true labels. In our grid search testing, $\alpha=0.1$ was found to be the best, showing that the signals are vastly different from each other.

$$L = \alpha D_{KL}(T_{soft} || S_{soft}) + (1 - \alpha) CE(y, S_{pred})$$

$$\alpha = \text{mixing coefficient}$$

$$T_{soft} = \text{Teacher model predictions for new class}$$

$$S_{soft} = \text{Student model predictions for new class}$$

$$CE(y, S_{pred}) = \text{Cross Entropy loss between the ground truth and student prediction}$$

Learning incrementally shows a small modest improvement in peak accuracy of ~84%. While inference time is the same, the model can learn new unknown signals without degradation.

General Adversarial Network is implemented to also increase accuracy by creating additional training samples. In general, GANs are difficult to engineer. Our model uses an ACGAN style network where the GAN is conditioned

with the class, so the discriminator learns to predict real/fake as well as the modulation class. Generating

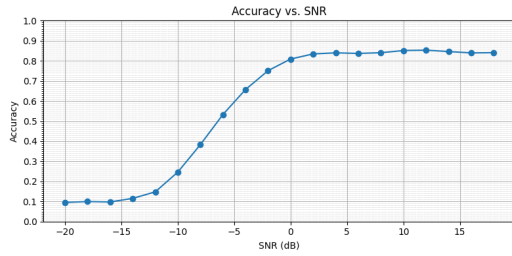


Figure 8: SNR Accuracy of OSIL

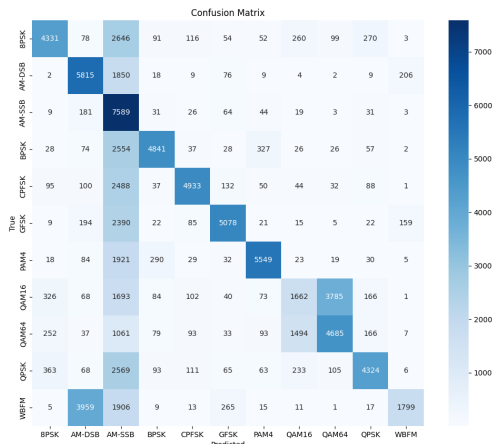


Figure 9: Confusion Matrix of OSIL

additional samples with low SNRs makes no sense and did not perform well. The GAN is trained with SNR of 18dB only to create the most realistic samples. GAN's inherently will add noise anyways. Another challenge for GANs is while they do start to stabilize their losses, visual checks are necessary to determine when training is at its best. The discriminator has dual outputs and therefore a dual loss structure. Binary cross entropy is used to determine if the generated sample is real/fake and categorical cross entropy is used to predict the softmax of the modulation classes.

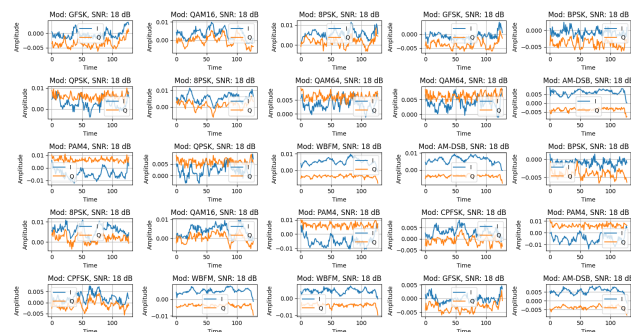


Figure 10: GAN Output after 90 epochs

The generator is trained with the discriminator, with the discriminator weights frozen, and therefore also has the same dual loss functions. The models were inspired by the work of Brownlee[8]. To not contaminate the test set, the GAN was trained on 6600 training samples which included all 11 classes. After about 90 epochs, the losses start to level out and visual inspection of the data is carried out to avoid overfitting. The GAN has created imperfect data, Figure 10, but it has learned certain subtle details as can be seen in separation of the I/Q channels for WBFM and AM-DSB. However, phase shifts in BPSK/QPSK are more difficult for the GAN to generate. With experimentation, it is found that an ~1% sample ratio of GAN generated vs original dataset samples is ideal for our model to achieve a higher peak accuracy of ~92%.

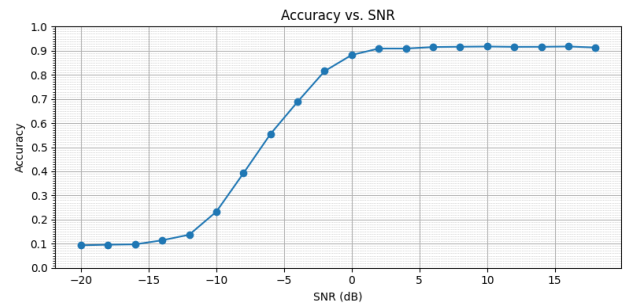


Figure 11: GAN SNR Accuracy Plot

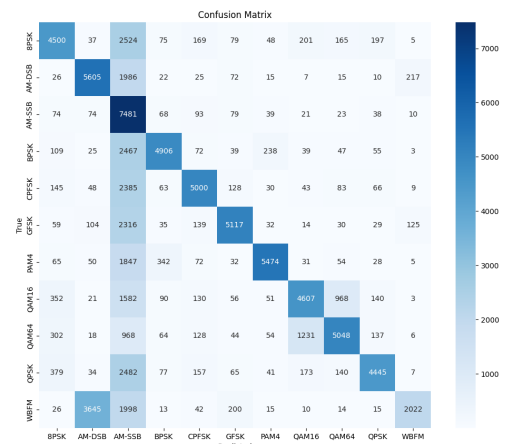


Figure 12: GAN Confusion Matrix

Dual Head model consists of two branches that get concatenated together like a y-network. The idea is to apply domain knowledge that the LSTM RNN network may not pick up to improve performance. One branch is the RNN base model (Figure 4) while the other is a convolutional neural network (CNN). The CNN's inputs are images using Fast Fourier Transforms (FFT) of the signal (Figure 13). The CNN layer is better suited for an

image and consists of 4 layers with 64x128 pixel images. The two branches were then concatenated with global average pooling followed by a dense layer with 128 units and ReLU activation. We added a 0.3% dropout layer and then a SoftMax layer for final classification. The complexity of this model and maxing out the 16GB VRAM on the GPU delayed further improvement of this model. We were able to achieve a peak accuracy ~73%, which proves the ability of this style model to have meaningful results.

```

Classification Report:
      precision    recall  f1-score   support

      8PSK         0.77      0.55      0.65      8000
      AM-DSB        0.54      0.72      0.62      8000
      AM-SSB        0.28      0.88      0.42      8000
      BPSK          0.83      0.61      0.70      8000
      CPFSK         0.84      0.62      0.71      8000
      GFSK          0.82      0.65      0.72      8000
      PAM4          0.88      0.70      0.78      8000
      QAM16         0.70      0.62      0.66      8000
      QAM64         0.82      0.61      0.70      8000
      QPSK          0.83      0.56      0.67      8000
      WBFM         0.69      0.28      0.39      8000

 accuracy          0.73      0.62      0.64     88000
  macro avg        0.73      0.62      0.64     88000
  weighted avg     0.73      0.62      0.64     88000

Done! 1745725033.459633
Total Training Time: 1016.72 seconds
Total Predicting Time: 7.45 seconds
Total Time: 1032.27 seconds

```

Figure 14: GAN Classification report

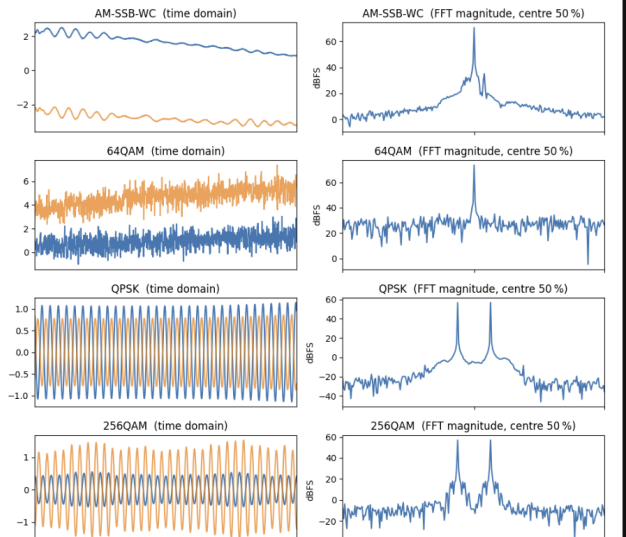


Figure 13 : FFT of RML2018 Dataset

Larger dataset can also be addressed by using the RML2018.01A dataset with over 2.5M samples across 24 modulation classes. Only 5 of them overlap with the RML2016.10A dataset, so we first look at the effects on the model independently. The approach for this was to use the same LSTM RNN model (Figure 4) with a 1024 input size

instead of 128. The model was otherwise kept the same. This is to accommodate for the higher temporal resolution. Training was done for 250 epochs. The initial dropout layer of 50% allowed us to prevent overfitting as seen in Figure 15 with a peak accuracy 85%. It's not possible to compare these two datasets equivalently, but we can infer that the model is strong for more data and more modulations. In efforts pseudo-test this, a cross dataset model was developed by grouping similar signal calcifications from the RML2016 set to the RML2018 set (e.g. AM-SSB -> AM-SSB-SC, AM-DSB -> AM-DSB-SC). We transferred the RML2016 model and then froze all the top layers except for the final softmax. This was trained on the RML2018 dataset for 25 epochs and 24 classes. Then the entire model was fully trained for another 200 epochs. The RML2016 dataset was input for prediction of the 11 classes. This gave us a peak accuracy of ~74%, showing again the generalization and ability to accurately detect new and novel modulations.

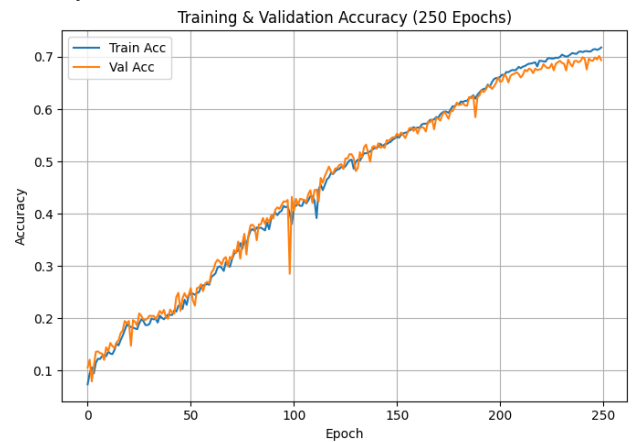


Figure 15: RML2018.01A LSTM RNN Accuracy Plot

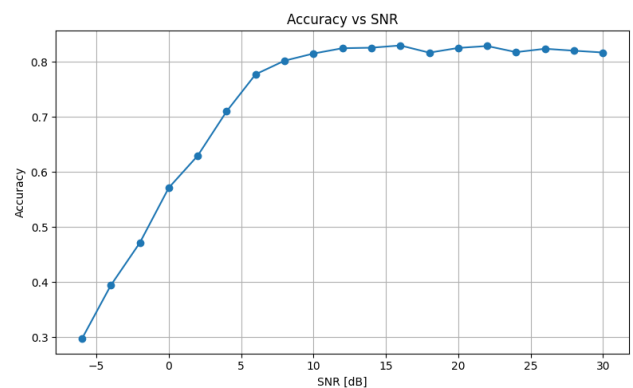


Figure 16: RML2018.01A SNR Accuracy Plot

4. Conclusion and Results

The best model that was reproducible was one by Hou et al. called the IQGMCL [6]. The authors address the

temporal nature with a bi-directional LSTM and add two inputs with temporal aspects of the individual I/Q signals looking at 50 points at a time. This model provided a peak accuracy of ~93% in the paper, however when reproduced, we got 91.3%. Inference time too 117 us per sample. Our best model with the GAN and additional dataset has a peak accuracy of ~92% and 86 us inference time per sample, an over 36% increase in prediction time. Our hypothesis that sample set is the largest contributor to accuracy is validated. Furthermore, data that has low (< 0 dB) SNR is expected to be intractable to predict and therefore these models are at diminishing returns for accuracy. All models were trained on a NVIDIA L40S GPU on the Virginia Tech ARC falcon cluster. Source code can be found on our GitHubs².

Model	Peak Acc	Inference Time per Sample
Base	83%	87 us
OSIL	84%	88 us
GAN	92%	85 us
Dual Head*	73%	n/a
RML2018 Cross*	74%	n/a
IQGMCL [6]	91-93%	117 us

Table 1: Model Summary (Model tested on RML2016 dataset unless marked. * = tested on RML2018 dataset)

Table 1 above summarized the results and shows that our model can perform state of the art predictions.

5. Discussion and Future Thoughts

A major strength of the work is the ability of the model to infer new signals using OSIL without forgetting older signals. This is critical in the real world as new modulations are constantly being created and evolved for defense purposes. Second, the model is able to infer in near real time with relatively low resolution data, to allow for application in the field.

The temporal nature of the signal, with its rapid shifts in phase, amplitude and other properties pose a challenge to traditional neural networks. Time-series data contain fine features that are not mapped perfectly to images and the advancements of convolutional networks. Future improvements include converting a time signal into a gram angular field (GAF), which in turn converts a time-series sample into an image. The use of a number of image based models like ResNet or AlexNet are now available for testing. Work is similar to that of Bai et al. but has not been applied to the RML201X datasets [9]. Another future

work is to use attention based models that not only look at the time-series data most adjacent to the data point (like LSTMs), but at the entire sample itself. The radio signal data can draw many similarities to natural language, even in the aspects where different signal shifts can indicate the same modulation, similar to synonym of words. Another set of OSIL models that was researched and successfully used on the CIFAR dataset is inspired by the works of Rebuffi et al. called “iCaRL” [10]. This model uses replay buffers to fill in imbalanced datasets and Mahalanobis distance to threshold unknown classes.

We hope this research encourages healthy dynamic spectrum access, interference mitigation and defense related signal intelligence.

6. Contributions

Contributions of this in depth study were broken out by experiments. Both team members worked on prior work to create the framework as previous coursework.

Paras used the RML2016 dataset to reproduce prior work, implement and review the general adversarial network and test open set incremental learning to prove the unknown signal challenge. This report was independently written by Paras, including the analysis and conclusion.

Jacob was a key contributor to the prior work and the RNN network that built the foundation of these experiments. Contributions include taking on a large and complex RML2018 dataset to ensure RNN model adherence. This included inspecting and understanding the nuances of the larger dataset. Further, Jacob was responsible for implementing and testing the dual head model as well as applying domain knowledge to select the engineered FFT features.

Both project members contributed equally to collaborate domain as well as deep learning knowledge to creatively find novel approaches to improve performance. This system level engineering is critical to the successful outcome of the report.

²<https://github.com/gawdygoda/ML-wireless-signal-classification/tree/main/DeepLearning>

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